MATH2241 – Introduction to mathematical analysis (1st semester 2526)

Tutorial 8 Solution

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1. Determine the biggest term of the sequence given by

   (a) $f_n = \dfrac{\sqrt{n}}{n + 2241}$

   (b) $g_n = \frac{n^2}{n^3+200}$

   Solution.

   (a) The function $f(x) = \dfrac{\sqrt{x}}{x + 2241}$ with $f'(x) = \dfrac{2241 - x}{2\sqrt{x}(x + 2241)^2}$ has its maximum at the point $x = 2241$, because $f' > 0$ for $x < 2241$ (the function is increasing), and $f' < 0$ for $x > 2241$ (the function is decreasing). So $n = 2241$.

   (b) The function $f(x) = \dfrac{x^2}{x^3 + 200}$ has a critical point at $x = \sqrt[3]{400} \approx 7.37$. $f(7) > f(8)$, it follows that $n = 7$.

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2. Prove the uniform continuity of the following functions by definition.

   (a) $f(x) = x^2 - 3x - 1$, $x \in [3, 6]$

   (b) $f(x) = 2x - 1$, $x \in \mathbb{R}$

   Solution.

   (a)

$$\begin{aligned} |f(x) - f(y)| &= |(x^2 - 3x - 1) - (y^2 - 3y - 1)| \\ &= |x - y| \cdot |x + y - 3| \\ &= |x - y| \cdot (|x| + |y| + |3|) \\ &\le 15|x - y| < \varepsilon \qquad \text{if } |x - y| < \frac{\varepsilon}{15} \end{aligned}$$

   $\forall \varepsilon > 0$, take $\delta = \dfrac{\varepsilon}{15}$, then $\forall x, y \in [3, 6]$, $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$.

(b)

$$\begin{aligned} |f(x) - f(y)| &= |(2x-1) - (2y-1)| \\ &= 2|x-y| < \varepsilon \qquad \text{if } |x-y| < \frac{\varepsilon}{2} \end{aligned}$$

$\forall \varepsilon > 0$, take $\delta = \dfrac{\varepsilon}{2}$, then $\forall x, y \in \mathbb{R}$, $|x-y| < \delta \implies |f(x) - f(y)| < \varepsilon$.

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3. Prove that the function $f(x) = \frac{|\sin x|}{x}$, $x \neq 0$, is uniformly continuous on each of the intervals $(-1, 0)$ and $(0, 1)$, but not on their union $(-1, 0) \cup (0, 1)$.

Solution.

The function f equals on the interval $(0, 1)$ to the function

$$F_1(x) = \begin{cases} 1 & \text{if } x = 0 \\ f(x) & \text{if } x \in (0, 1] \end{cases}$$

F_1 is continuous on the closed interval $[0, 1]$, hence it is uniformly continuous there. This also gives the uniform continuity of f on the interval $(0, 1)$. Analogously, f is also uniformly continuous on $(-1, 0)$.

However, f is not uniformly continuous on the union $A := (-1, 0 \cup (0, 1)$. Let $x_n = -\frac{1}{n+1}$ and $x'_n = \frac{1}{n+1}$ for $n \in \mathbb{N}$. For any $\delta > 0$, there is $n = n_\delta$ s.t.

$$|x_n - x'_n| = \frac{2}{n_1} < \delta.$$

But for $n \in \mathbb{N}$, f is decreasing on $(0, \frac{\pi}{2})$, take $\varepsilon = 1$

$$|f(x_n) - f(x'_n)| = 2(n+1)\left|\sin\frac{1}{n+1}\right| \geq 2 \cdot \frac{2}{\pi}\left|\sin\frac{\pi}{2}\right| = \frac{4}{\pi} > 1 = \varepsilon$$

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4. Use the Mean Value Theorem to show that $|\sin w - \sin z| \leq |w - z|$ for $0 \leq z < w \leq 2\pi$.

Solution.

$f = \sin x$ is continuous and differentiable. By MVT, there is $c \in (w, z)$ s.t.

$$\begin{aligned} \left|\frac{\sin w - \sin z}{w - z}\right| &= |f'(c)| = |\cos c| \leq 1 \\ |\sin w - \sin z| &\leq |w - z| \end{aligned}$$

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5. Suppose $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b). Suppose $f(a) = 0$ and $|f'(x)| \leq A|f(x)|$ for a constant A.

   (a) Prove that $f(x) = 0$ on $\left[a, a + \dfrac{1}{2A}\right]$.

   (b) Prove that $f(x) = 0$ on the whole interval $[a, b]$.

   Solution.

   (a) MVT on the interval $[a, x]$ for $x \in [a, a + \frac{1}{2A}] \cap [a, b]$, $\exists c_1 \in (a, x)$ s.t.

$$\frac{|f(x) - f(a)|}{|a + \frac{1}{2A} - a|} \leq \frac{|f(x) - f(a)|}{|x - a|} = f'(c_1) \leq A|f(c_1)|$$
$$2A|f(x)| \leq A|f(c_1)|$$
$$|f(x)| \leq \frac{|f(c_1)|}{2}$$

   Then, MVT again on the interval $[a, c_1]$, $\exists c_2 \in (a, x)$ s.t. $|f(x)| \leq \dfrac{|f(c_1)|}{2} \leq \dfrac{|f(c_2)|}{2^2}$. Inductively, $|f(x)| \leq \dfrac{|f(c_n)|}{2^n}$. f is continuous on $[a, b]$, it is bounded, $\exists M > 0$ s.t. $|f| < M$ on $[a, b]$.
   Now, $\forall \varepsilon > 0$, $\forall x \in \left[a, a + \frac{1}{2A}\right]$, take n large enough that $\dfrac{1}{2^n} < \dfrac{\varepsilon}{M}$, we have

$$|f(x)| \leq \frac{|f(c_n)|}{2^n} < \frac{M}{2^n} < \varepsilon$$

   So, $f(x) = 0$ on $\left[a, a + \dfrac{1}{2A}\right]$.

   (b) Inductively using (a) and $\frac{1}{2A} > 0$, $f(x) = 0$ on $\left[a, a + \dfrac{n}{2A}\right]$ and then the whole interval $[a, b]$.

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— *End* —